

SYZRAN, Russian Federation

WMO: 279830

Lat: 53.183N

Lon: 48.400E

Elev: 103

StdP: 100.09

Time Zone: 4.00 (E04)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.3	-23.2	-29.3	0.3	-26.2	-26.4	0.3	-22.8	8.3	-6.3	7.2	-9.1	1.7	260	0.555

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	31.9	19.6	29.8	19.2	28.0	18.6	21.1	28.3	20.3	27.5	19.6	26.3	3.0	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	13.7	24.1	17.9	13.0	23.4	17.1	12.3	22.5	61.6	28.2	58.8	27.5	56.2	26.5	23.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.8	6.8	6.1	DB	-29.9	34.5	3.4	2.0	-32.4	35.9	-34.4	37.0	-36.3	38.1	-38.8	39.6	(4)
(5)			WB	-29.9	22.7	3.2	0.6	-32.3	23.1	-34.2	23.4	-36.0	23.8	-38.3	24.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	5.8	-10.1	-9.8	-3.3	7.0	14.2	18.9	21.2	19.3	13.5	6.1	-1.2	-7.4	(6)
(7)		DBStd	12.38	6.77	6.86	5.08	5.30	4.42	4.10	3.61	3.98	4.18	4.22	5.34	6.61	(7)
(8)		HDD10.0	2747	624	555	413	121	14	1	0	0	12	133	335	539	(8)
(9)		HDD18.3	4829	883	788	672	341	138	42	14	37	154	378	585	798	(9)
(10)		CDD10.0	1209	0	0	0	31	146	268	346	288	118	13	0	0	(10)
(11)		CDD18.3	249	0	0	0	1	11	59	102	66	10	0	0	0	(11)
(12)		CDH23.3	2209	0	0	0	11	126	525	898	572	78	0	0	0	(12)
(13)		CDH26.7	732	0	0	0	1	17	173	314	213	15	0	0	0	(13)
(14)	Wind	WSAvg	2.9	2.8	2.9	3.0	3.2	3.2	3.0	2.7	2.7	2.7	2.9	3.0	2.9	(14)
(15)	Precipitation	PrecAvg	455	31	23	19	34	32	57	62	52	42	38	37	32	(15)
(16)		PrecMax	641	61	59	51	94	98	106	154	198	128	81	101	75	(16)
(17)		PrecMin	288	4	0	0	0	0	10	2	0	0	0	9	8	(17)
(18)		PrecStd	87	14	16	14	23	23	23	40	38	34	22	21	17	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.7	2.9	10.5	24.9	28.5	34.3	35.3	35.1	28.6	19.4	11.1	4.1	(19)
(20)			MCWB	1.7	1.2	6.2	13.2	16.1	19.5	19.9	19.4	16.5	12.1	9.5	3.2	(20)
(21)		2%	DB	1.6	1.9	6.4	21.5	26.5	30.8	31.8	31.4	25.5	16.6	8.9	2.6	(21)
(22)			MCWB	0.5	0.7	3.3	12.0	15.7	19.1	20.0	19.3	15.4	11.0	7.3	1.4	(22)
(23)		5%	DB	0.5	0.9	4.2	18.5	24.4	28.4	29.8	28.9	22.9	14.3	6.9	1.6	(23)
(24)			MCWB	-0.3	0.0	1.9	10.7	15.2	18.8	19.7	18.9	15.0	10.3	5.7	0.6	(24)
(25)		10%	DB	-0.6	-0.2	2.6	15.7	22.2	26.2	28.0	26.3	20.2	12.4	5.3	0.7	(25)
(26)			MCWB	-1.4	-1.1	0.9	9.3	13.9	17.9	19.3	18.1	14.0	9.2	4.2	-0.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.7	1.6	6.6	14.5	18.1	21.5	22.3	21.5	18.2	13.4	9.9	3.6	(27)
(28)			MCDWB	2.3	2.3	10.5	21.6	24.7	28.9	29.3	29.1	24.8	16.4	10.9	4.1	(28)
(29)		2%	WB	0.6	0.8	3.6	12.9	17.0	20.3	21.5	20.4	16.8	11.9	7.6	1.6	(29)
(30)			MCDWB	1.3	1.7	5.7	20.1	24.2	27.9	28.4	28.4	22.7	15.4	8.7	2.2	(30)
(31)		5%	WB	-0.2	0.1	2.1	11.1	15.9	19.5	20.8	19.6	15.7	10.6	5.7	0.7	(31)
(32)			MCDWB	0.5	0.9	3.8	17.8	22.6	26.7	27.5	27.3	20.9	13.8	6.7	1.4	(32)
(33)		10%	WB	-1.4	-1.1	1.1	9.6	14.8	18.6	20.0	18.7	14.5	9.5	4.3	0.0	(33)
(34)			MCDWB	-0.6	-0.2	2.4	15.2	21.2	25.0	26.4	25.2	19.6	12.2	5.2	0.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.9	6.8	7.3	9.6	10.9	10.5	10.3	10.4	9.6	7.0	4.7	5.2	(35)
(36)			MCDBR	4.4	5.2	8.1	14.0	14.5	13.9	13.4	14.7	13.7	11.1	6.0	4.1	(36)
(37)			MCWBR	4.3	4.6	5.7	6.9	6.2	5.4	4.8	5.3	5.8	6.3	5.0	3.9	(37)
(38)		5% WB	MCDWB	4.4	4.8	7.2	13.2	12.6	11.6	11.6	12.5	11.9	9.5	5.3	3.9	(38)
(39)			MCWBR	4.3	4.5	5.2	7.0	6.0	5.0	4.9	5.3	5.8	5.9	4.8	3.9	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.297	0.327	0.367	0.431	0.404	0.393	0.407	0.429	0.381	0.353	0.326	0.297	(40)	
(41)		taud	2.082	2.039	2.033	2.088	2.286	2.363	2.321	2.244	2.350	2.340	2.238	2.101	(41)	
(42)		Ebn at Noon	641	743	795	794	845	858	837	791	789	725	616	560	(42)	
(43)		Edh at Noon	73	105	132	143	123	116	119	121	97	79	64	60	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.88	1.93	3.37	4.35	5.62	6.08	6.01	4.90	3.23	1.64	0.70	0.48	(44)	
(45)		RadStd	0.15	0.22	0.39	0.41	0.38	0.57	0.46	0.41	0.44	0.27	0.11	0.14	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	+1.63	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (4 neighbors)		N/A	N/A	N/A	+1.12	N/A	N/A	N/A	N/A	+93	N/A

Nomenclature: See separate page